1. Power the circuit with 5V DC.
2. Press S1 (up) or S2 (down) until LED2 to LED5 all glow, indicating 1111.
3. Connect AP16 to AP16′, 5V to 5V′ and GND to GND′, and connect APPL16 to CON16.
4. Connect the 230V AC supply to CON17.
5. Switch S3 and S4 on. This selects output S15 of IC3, which corresponds to APPL16.
6. IC3 pulls its S15 output (pin 17) low, which turns T16 on and energises RL16.
7. The relay contacts connect AP16 to AP16′ and switch APPL16 on.
8. To switch APPL16 off, turn S3 or S4 off. S4 can be used to switch off any appliance at any time.

Safety note. The 230V AC section is hazardous. Use proper insulation, protection, rated components, a secure enclosure, and earthing. Mains wiring and testing must be done by a qualified person.

Troubleshooting

If a channel does not work, check the 5V supply, IC orientation, push-button connections, and the wiring around the counter and decoder. Also inspect the BC557 transistors, relay coils, flyback diodes, and indicator LEDs. Test each channel individually and check that the counter and decoder outputs change correctly.

Applications

This circuit can be adapted for lab equipment control, demonstration panels, and digital electronics teaching projects. EFY

S.C. Dwivedi is an electronics enthusiast and circuit designer at EFY

BONUS CONTENT ONLINE!

On our website **ElectronicsForU.com** you will find additional content for this project, which includes:
1. Demo video: See the project working
2. PCB layouts

EFY TESTED Ashwini K. Sinha

MONITOR CONCRETE STRENGTH Wirelessly

Two beams, one pour day, very different strengths? Calendar age hides the real story. In just '5 hours' of build time, you can assemble a 'medium' complexity system under ₹2500: an ESP32 and three embedded sensors that log concrete's temperature history and calculate its maturity—wirelessly!

ASHWINI KUMAR SINHA

Two beams poured on the same day can end up with very different strengths. Calendar age hides what matters: temperature history. Concrete does not gain strength at a constant rate because hydration is strongly driven by temperature, so two elements of the same chronological age can be at very different stages of development.

This ESP32-based system embeds three DS18B20 sensors within a concrete pour and records that history. It continuously calculates the average temperature, the temperature difference between locations, Nurse–Saul maturity, and Arrhenius equivalent age, all without an internet connection.

The ESP32 creates its own Wi-Fi access point, named 'Concrete-Maturity'. A phone, tablet, or laptop connects directly to it, and opening 192.168.4.1 in a browser shows a live dashboard of temperatures and maturity values. The dashboard doubles as a data logger: readings are timestamped and stored in the browser while recording, and you can export a comma-separated values (CSV) file, save the temperature graph as an image, or generate a self-contained hypertext markup language (HTML) engineering report with measured ranges, maturity values, equivalent age, the graph,



Fig. 1: Author's prototype